MIL-R-46903B (MU)
16 AUGUST 1965
SUPERSEDING
MIL-R-46903A (MU)
20 APRIL 1964

MILITARY SPECIFICATION

RELAY, EXPLOSIVE, XM10 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the Loading, Assembling and Packing for one type of explosive relay designated as Relay, Explosive, XM10.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

8864437 — Relay, XM10, Loading Assembly.

7548347 — Box, Set-up, Packing, Ammunition for Relays and Primers.

7548388 —Box, Packing, Ammunition for Relay M4, M5, M7, XM10, or Primer M42.

7548389 — Box, Set-up, Packing, Ammunition Inner, for Relays and Primers.

FSC 1375

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PUBLICATIONS

ARMY

IEL–8864437 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Relay assemblies. The loaded relay assemblies shall comply with all requirements specified in Drawing (dwg.) 8864437 and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide. The moisture contents of lead azide at the loading station at the time of loading the relays shall not exceed 0.50 percent when tested as specified in 4.4.1.

3.4 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be clean and free of burrs, surface defects, sharp edges, cracks, dirt, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define terms used herein. The provisions of MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government specifications or commercial standards referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's

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authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 *Submission.* Prior to initiation of regular production, the contractor shall submit a sample of 25 assemblies to a Government approved facility designated by the contracting officer (see 6.4) for testing in accordance with provisions of 4.2.2. All parts and materials shall be the same as will be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract.

4.2.2 *Inspections to be performed.* The assemblies will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly fails to comply with any or all of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of loaded relays produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot interfix number. Lots shall be submitted for inspection in accordance with MIL–STD–105 (or MIL–STD–1235, when applicable). Each lot shall contain:

(a) Parts of one lot interfix number from one manufacturer.

(b) Lead azide from not more than one lot.

4.3.2 *Examination.* Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Assembly, less disc relay (dwg 8864436) and washer relay (dwg 8844437) and prior to crimping. (see dwg 8864437)

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:	None defined.		
Major:	AQL 0.40 percent	(Note)	
101.	Weight of upper charge	Balance	01001
102.	Weight of lower charge	Balance	01002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	01003

4.3.2.2 Assembly (see dwg. 8864437)

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Explosive exposed	Visual	02001
2.	Cup cracked or split	Visual	02002

Note. When automatic weighing equipment is utilized, the provisions of paragraphs 4.3.3.3 and 4.4.2 shall apply.

Major:	AQL 0.40 percent		
101.	Charge missing	Visual	02003
102.	Split or torn relay disc	Visual	02004
Minor:	AQL 2.5 percent		
201.	Outside diameter	Gage	02005
202.	Total length	Gage	02006
203	Under side of crimped metal fails to contact disc a full 360 degrees	Visual	02007
204.	Any portion of washer hole obstructed by crimp	Visual	02008
205.	Component missing	Visual	02009
206.	Color incorrect or missing	Visual	02010
207.	Evidence of poor workmanship (see 3.4)	Visual	02011

4.3.2.3 Box, set-up, inner (see dwg. 7548389)

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Number of relays incorrect	Visual	03001

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Categories	Defects	Method of inspection	Code No.
Minor:	AQL 1.5 percent		
201.	Any packing material missing	Visual	03002
202.	Spacer damaged to the extent that the relays touch each other	Visual	03003
203.	Label missing	Visual	03004
204.	Marking misleading or unidentifiable	Visual	03005

4.3.2.4 Box, set-up, packing, ammunition, outer (see dwg. 7548347)

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.5 percent		
201.	Contents loose	Manual	04001
202.	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal box	Visual	04002
203.	Box torn, cut or punctured	Visual	04003
204.	Label missing	Visual	04004
205.	Marking misleading or unidentifiable	Visual	04005

4.3.2.5 Sealed barrier bag (see dwg. 7548347)

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Seal improper or incomplete	Visual	05001
102.	Bag punctured, torn or cut	Visual	05002
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable	Visual	05003

4.3.2.6 Box, packing (see dwg. 7548388)

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Box broken, board loose or protruding nail	Visual	06001
102.	Strapping missing, broken or loose	Visual-Manual	06002

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Categories	Defects	Method of inspection	Code No.
Minor:	AQL 1.5 percent		
201.	Contents loose	Visual-Manual	06003
202.	Strapping improperly engaged or misleading	Visual	06004
203.	Marking misleading or unidentifiable	Visual	06005

4.3.3 *Testing.*

4.3.3.1 *Moisture content of lead azide (see 3.3) — Major defect — Code No. 07001.* The moisture content of each lot shall be determined at the time of loading. A sample of 2 grams shall be selected and tested as specified in 4.4.1. If the moisture content is in excess of the percentage specified, and loading has not begun, the lot of lead azide shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.4.1. If loading has begun, action on the unloaded lead azide shall be as specified above and all relays loaded with the questionable lead azide shall be rejected.

4.3.3.2 *Heat seal test of sealed bag (see dwg. 7548347) Major defect — Code No. 05004.* Sampling, acceptance criteria and test procedure for this test shall be as prescribed in MIL–P–116 for the Heat Seal Test.

4.3.3.3 *Weight of charges (see dwg. 8864437) — Major defect.* Twenty samples of each charge (upper & lower) shall be selected as representative of each eight hours production. If one or more charges fails to meet the applicable weight requirement the eight hours production represented shall be rejected (see para 4.4.2).

4.3.4 *Inspection Equipment.* Index of Inspection Equipment List Number IEL–8864437 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 **Test methods and procedures.**

4.4.1 *Determination of moisture content of lead azide.* A dried aluminum dish 2 inches in diameter shall be covered with a sheet of polytetrofluoroethylene 2½ inches square and sufficiently flat to form a tight

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cover and the tare weight determined. An accurately weighed portion of approximately two grams of the sample shall be transferred to an oven and dried at 95 degrees centigrade plus or minus 2 degrees Centigrade for 2 hours. The cover shall be placed in a desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The polytetrofluoroethylene cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated as the percentage of moisture content in the lead azide to determine compliance with 3.3 as follows:

Percentage of Moisture = (A minus B) times 100 divided by A.

where:

A = The weight of the sample before drying.

B = The weight of the sample after drying.

4.4.2 *Weight of charges.* Two hundred empty relay cups shall be selected, weighed, and the average cup weight determined. Samples representing each charge (upper and lower) shall be obtained from the automatic weighing equipment and manually weighed. Determination of the exactness of the automatic weighing equipment shall be ascertained by deletion of the average empty cup weight from the total weight of sample charge. Equipment shall be in accordance with 4.3.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level C. — Unit packaging.* The relays shall be packaged in accordance with dwg. 7548389.

5.1.2 *Intermediate packaging.* Unit packages shall be packaged in outer containers in accordance with dwg. 7548347.

5.2 Packing.

5.2.1 *Level C.* The relays shall be packed in accordance with dwg. 7548388.

5.3 Marking. Unit, intermediate and exterior containers shall be marked in accordance with dwgs. 7548389, 7548347 and 7548388.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Ammunition data cards shall be prepared for each lot in accordance with the information specified in MIL–STD–1167.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the number column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equip-

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ment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as Contractor Design in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL-D-45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all informa-

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tion necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black line on white background and shall provide a legible print produced via the 35MM microfilm process. Contractor shall submit one copy of the final design as a rolled set or flat set.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of initial production samples. Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN; SMUPA–NI, Dover, N. J.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1375–A956

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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